

Differential Calculus

*Home work sheet # 11

Conic Section

3–6 Sketch the parabola, and label the focus, vertex, and directrix. ■

3. (a) $y^2 = 4x$ (b) $x^2 = -8y$
4. (a) $y^2 = -10x$ (b) $x^2 = 4y$
5. (a) $(y - 1)^2 = -12(x + 4)$ (b) $(x - 1)^2 = 2(y - \frac{1}{2})$
6. (a) $y^2 - 6y - 2x + 1 = 0$ (b) $y = 4x^2 + 8x + 5$

7–10 Sketch the ellipse, and label the foci, vertices, and ends of the minor axis. ■

7. (a) $\frac{x^2}{16} + \frac{y^2}{9} = 1$ (b) $9x^2 + y^2 = 9$
8. (a) $\frac{x^2}{25} + \frac{y^2}{4} = 1$ (b) $4x^2 + y^2 = 36$
9. (a) $(x + 3)^2 + 4(y - 5)^2 = 16$
(b) $\frac{1}{4}x^2 + \frac{1}{9}(y + 2)^2 - 1 = 0$
10. (a) $9x^2 + 4y^2 - 18x + 24y + 9 = 0$
(b) $5x^2 + 9y^2 + 20x - 54y = -56$

11–14 Sketch the hyperbola, and label the vertices, foci, and asymptotes. ■

11. (a) $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (b) $9y^2 - x^2 = 36$
12. (a) $\frac{y^2}{9} - \frac{x^2}{25} = 1$ (b) $16x^2 - 25y^2 = 400$
13. (a) $\frac{(y + 4)^2}{3} - \frac{(x - 2)^2}{5} = 1$
(b) $16(x + 1)^2 - 8(y - 3)^2 = 16$
14. (a) $x^2 - 4y^2 + 2x + 8y - 7 = 0$
(b) $16x^2 - y^2 - 32x - 6y = 57$

15–18 Find an equation for the parabola that satisfies the given conditions. ■

15. (a) Vertex (0, 0); focus (3, 0).
(b) Vertex (0, 0); directrix $y = \frac{1}{4}$.
16. (a) Focus (6, 0); directrix $x = -6$.
(b) Focus (1, 1); directrix $y = -2$.
17. Axis $y = 0$; passes through (3, 2) and (2, $-\sqrt{2}$).

**These problems are for the students only as home work. Search the reference books for more examples.*